SNAPDRAGON AI LAB BUILD & PRESENT CHALLENGE

Rakshak Edge

A private, on-device social-engineering firewall for Snapdragon-powered PCs

PAUSE THE SCAM BEFORE THE TAP

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Executive summary

Rakshak Edge analyzes suspicious messages, payment requests, QR payloads, and links before a user acts. It detects urgency, impersonation, credential theft, risky URL structure, and Indian payment signals. A trained local semantic classifier catches linkless social-engineering patterns, while the evidence layer explains the result and recommends a safer verification path. The complete build works fully offline with no account, telemetry, or cloud upload.

> **The product protects the decision point: the few seconds before a person clicks a link, shares an OTP, or approves a payment.**

Problem

Social-engineering scams exploit context, urgency, and trusted identities. A link scanner alone misses scams such as digital-arrest calls or requests from a compromised contact. A cloud-only checker introduces another privacy decision because the user must upload the very message, account detail, or payment request they distrust.

Solution

Rakshak Edge accepts copied text from SMS, email, WhatsApp, a URL, or a QR decoder. It extracts security-relevant entities, scores visible manipulation patterns, and produces an evidence-first report. High-risk results begin with a clear stop action and direct the user to an independently trusted contact path.

Evaluation area	Evidence in this submission
Technical implementation	Working local API, trained semantic classifier, transparent corpus, entity extraction, URL analysis, calibrated rules, tests, and QNN upgrade adapter
Use case and innovation	Pre-action safety for link and non-link scams, Indian payment vocabulary, multilingual signals, and explainable guidance
Deployment and accessibility	Zero-dependency demo, CPU fallback, QNN production path, keyboard support, high contrast, and reduced motion
Presentation and documentation	Pitch deck, brief, live demo script, architecture, threat model, deployment guide, and clean repository

Technical implementation

The architecture uses a hybrid pipeline because security decisions need both semantic context and traceable evidence.

1	**Input boundary**	Loopback-only API, bounded payload, no server-side persistence
2	**Transparent analysis**	URL structure, UPI handles, phone numbers, amounts, credential requests, authority claims, urgency, and coercion
3	**Semantic model**	Trained word and character n-gram classifier for paraphrases and mixed language
4	**Evidence fusion**	Calibrated score, top reasons, extracted entities, and safe actions

Measured model and Snapdragon optimization

The checked-in classifier trains from 72 multilingual examples and achieves 94.44% accuracy on a fixed 18-message holdout. All six high-risk and all six low-risk examples are correct; one medium-risk example is missed and disclosed by the evaluation script. The Snapdragon production profile upgrades the compact reference model to a quantized encoder through ONNX Runtime's QNN Execution Provider. Static INT8 quantization reduces memory traffic while the transparent rule and evidence layer stays on CPU.

Holdout accuracy	Fixed holdout	Regression tests	Default cloud calls
94.44%	18 messages	7 passing	0

The holdout is intentionally small and transparent. It proves the end-to-end AI pipeline and reproducibility, not production-scale generalization. Target-device NPU validation remains an explicit acceptance check.

Safety and responsibility

Rakshak Edge reports risk indicators and safer actions. It does not claim that a message is proven fraudulent or safe. The UI uses textContent for user-derived values, the server constrains static-file paths, and the content security policy limits scripts and connections to the local origin. The threat model documents identity verification, malware sandboxing, and network reputation as out of scope.

Product experience

The interface uses a direct three-step mental model: paste, understand, act safely. Demo scenarios cover KYC and OTP theft, digital arrest, job-deposit fraud, and a legitimate security warning. Each result shows measured local latency, top evidence, and one primary action.

Scenario	Expected evidence	Primary response
KYC + OTP	Credential request, false urgency, look-alike SBI domain, UPI payment	Stop. Contact the bank through the official app or known number.
Boss request	99.8% high-risk semantic output plus voucher-payment evidence; no URL, OTP, or UPI	Call the finance head using the number already in your contacts.
Safe warning	Security-awareness context discounts quoted risk terms	No strong pattern found; continue to verify unexpected requests.

Run and evaluate

Start the product with **python app.py**, then open **http://127.0.0.1:8765**. Run the regression suite with **python -m unittest discover -s tests -v**. Reproduce the fixed holdout report with **python benchmarks/evaluate_model.py** and the latency harness with **python benchmarks/run_benchmark.py**.

Roadmap

The next milestone expands the consented multilingual corpus, calibrates confidence across scam families, adds Windows share-target integration, and validates a QNN INT8 encoder on a Snapdragon-powered HP PC. A later release can add opt-in federation that shares model updates without raw message content.

Key references

1. ONNX Runtime, Qualcomm QNN Execution Provider: https://onnxruntime.ai/docs/execution-providers/QNN-ExecutionProvider.html
2. Qualcomm AI Hub: https://aihub.qualcomm.com/
3. Snapdragon X Elite product brief: https://docs.qualcomm.com/bundle/publicresource/87-71417-1_REV_D_Snapdragon_X_Elite_Product_Brief.pdf

Repository package

The folder contains the working application, tests, benchmarks, architecture, threat model, Snapdragon deployment guide, demo script, submission copy, pitch deck, and this brief. Add the public GitHub URL to the Unstop form after pushing the repository.